DATA ANALYSIS FROM THE GERMAN CREDIT DATA SET

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INTRODUCTION

PROJECT GOAL: The aim of this project is to identify factors associated with loan default and to examine whether these factors have a statistically significant relationship with default status.
DATA SET: The dataset is the German Credit dataset (https://www.kaggle.com/datasets/rihabkaci99/credit-risk-dataset) contains loan applicant records used for **credit risk assessment**, which consists of 20 predictor variables describing the financial, personal, and socio-economic characteristics of loan applicants, along with one target variable indicating credit risk outcome. Each observation corresponds to a single applicant and includes information on checking and savings account balances, credit history, loan purpose and amount, employment length, installment rate as a percentage of income, housing status, number of existing credits, and property ownership. In addition, demographic attributes such as age, marital and personal status, job classification, dependents,

telephone availability, and foreign worker status are provided. The predictors comprise a combination of categorical and numerical features, allowing for comprehensive analysis of factors influencing creditworthiness. The target variable classifies applicants into good or bad credit, making the dataset well suited for supervised learning and credit scoring research.

TOOLS USED: Google Sheets , SQL.

DATA CLEANING

MISSING VALUES: There were no missing values
AGE GROUPS: Age values below 30 were grouped as "Young", values between 30 and 50 as "Middle", and values over 50 as "Senior".
RATE CATEGORY: Installment rates between 1 and 2 were grouped as "Low", rates equal to 3 as "Middle", and rates equal to 4 as "High".
EMPLOYMENT LENGTH CATEGORY: During the data preprocessing stage, an inconsistency was identified between the variables age and employment_length. Specifically, some observations indicated that individuals with relatively young ages (e.g., 20–23 years old) had employment experience exceeding the logically possible working years, assuming that employment typically begins at around 18 years of age.To facilitate quantitative analysis, the categorical variable employment_length was converted into a numeric variable using the midpoint of each interval (e.g., “1–4 years” → 2.5 years, “4–7 years” → 5.5 years). A new column named as **employment_length_num** is created in **first_edit** sheet including these values .
However, this transformation revealed cases where the estimated employment duration exceeded the maximum possible working years (age − 18).To preserve the original data while addressing logical inconsistencies, a corrected employment experience variable was created by applying an upper bound based on the maximum possible working years (age − 18). Additionally, a binary flag variable was introduced to indicate observations where the reported employment length exceeded the logical constraint. So, two new columns are created in **first_edit** sheet, named as **corrected_employment_length** and **flag_inconsistency.**
A histogram of the corrected employment length variable was examined to understand its distribution. Based on the observed concentration of values, the variable was re-categorized into meaningful experience groups representing different stages of employment. The corrected employment length variable is grouped as follows: values equal to 0 are classified as "Unemployed", values between 0.5 and 2 as "Very Short", values between 2.5 and 3 as "Short", values between 4 and 6 as "Medium", and values over 6 as "High".
OUTLIER HANDLING : The IQR method was used to explore the distribution of borrower age loan amount and loan duration. Extreme observations were identified but not removed because represent reasonable cases and sometimes connected with legitimate high-risk credit profiles they are important for predictive modeling in credit risk analysis. In a Google Sheet named "Outlier Check", a table was created containing all the necessary elements of the IQR method, such as Q1, Q3, IQR, upper bound, and lower bound. Another table was also created to check whether any individual value of the variables mentioned above meets the criteria to be characterized as an outlier.
OTHER: In order to focus the analysis on specific variables, 15 columns were deleted. The numerical variable representing loan amount, which was initially recorded in DM (German Mark), was

transformed into euros (€). The original numerical types were converted to types compatible with SQL. A check for duplicate values was performed, and no duplicates were found.
TOOLS: The Google Sheets web application and its features were used for the main data cleaning process. The SQL programming language was used to create categories for variables such as age, employment length, and installment rate. The main SQL statements used for queries at this stage were: CASE , WHEN , SELECT , FROM. For example : ***select ****, ***case*** ***when*** *age <30 then 'Young'* ***when*** *age between 30 and 50 then 'Middle' else 'senior' end as age_group* ***from*** *credit_risk* .

SQL ANALYSIS

FIRST QUERY:

SQL 1

```
select "default",
count(*) as cnt_default , round(count(*) * 100 / (select count(*) from view_groups) , 2 )
as percentage from view_groups
group by "default"
```

	default	cnt_default	percentage
1	BAD GREDIT	300	30.0
2	GOOD GREDIT	700	70.0

The first query was created to calculate the total **bad credit** rate in the database. As we can see, it is 30%. In other words, 300 out of 1,000 applications are considered **bad credit** cases or high-risk applications.

SECOND QUERY:

Database Structure | Browse Data | Edit Pragmas | Execute SQL

SQL 1

```sql
select age_group,
count(*) as total_cnt,
round(100.0*sum(case when "default" = 'BAD GREDIT' then 1 else 0 end) / count (*),2)
as percentage_bad_gredit
from view_groups
group by age_group
order by percentage_bad_gredit
```

	age_group	total_cnt	percentage_bad_gredit
1	Middle	516	25.58
2	senior	113	27.43
3	Young	371	36.93

From the second query, we can see the total number of applicants in each age group (total_cnt) and the corresponding **bad credit** percentage (percentage_bad_gredit).

THIRD QUERY :

This query first shows the number of applicants in each installment rate category out of a total of 1,000 applicants. It also shows the **bad credit** percentage (BAD_GREDIT_PERCENTAGE) for each category.

SQL 1

```sql
SELECT rate_category,
count(*) as total_count,
round(100.0*sum(case when "default" = 'BAD GREDIT' THEN 1 ELSE 0 END)/COUNT(*),2)
AS bad_gredit_percentage

from view_groups
group by rate_category
order by bad_gredit_percentage
```

	rate_category	total_count	bad_gredit_percentage
1	Low	136	25.0
2	Middle	388	27.58
3	High	476	33.4

FORTH QUERY

Database Structure | Browse Data | Edit Pragmas | Execute SQL

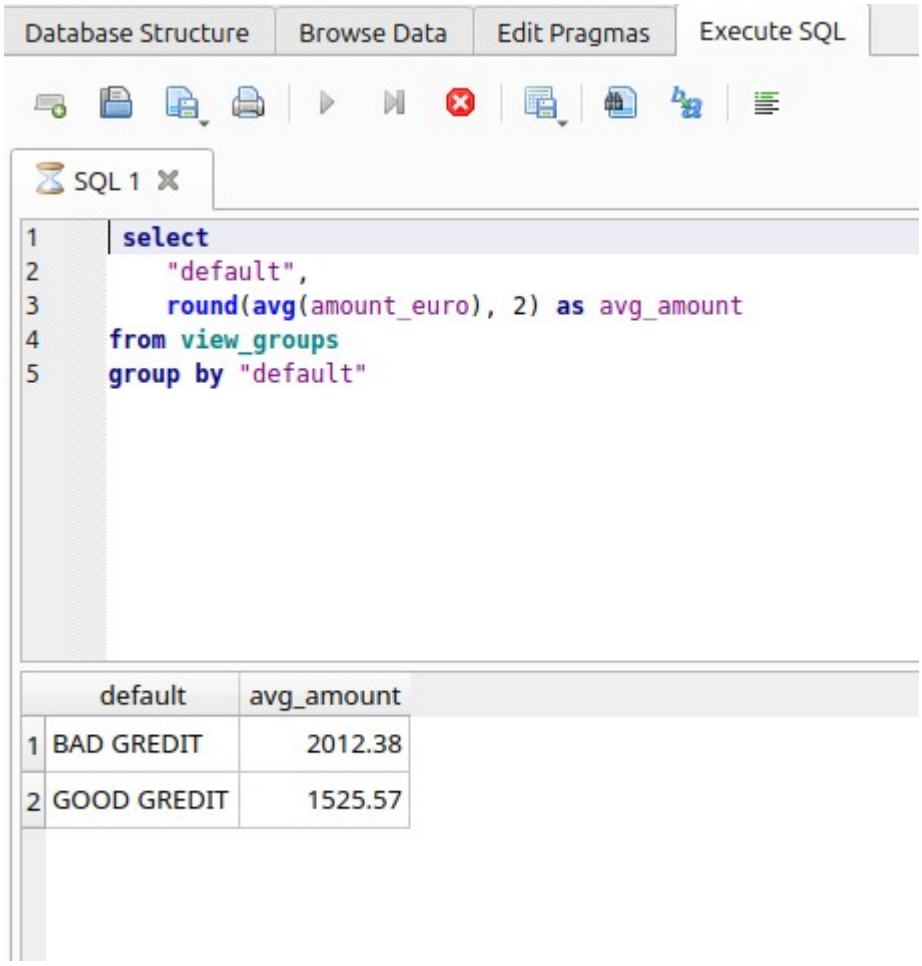
SQL 1

```sql
select employment_length_category,
count(*) as total_count,
round(100.0*sum(case when "default" = 'BAD GREDIT' THEN 1 ELSE 0 END)/COUNT(*),2)
AS BAD_GREDIT_PERCENTAGE
from view_groups
group by employment_length_category
ORDER BY BAD_GREDIT_PERCENTAGE
```

	employment_length_category	total_count	BAD_GREDIT_PERCENTAGE
1	medium	175	22.29
2	High	244	25.41
3	small	335	30.15
4	unemployed	62	37.1
5	very small	184	40.76

In this query, we can see the distribution of the total number of applicants across employment length categories. We can also observe that the **bad credit** percentage (BAD_GREDIT_PERCENTAGE) is significantly higher among the applicants with very short employment length (employment_ length_ category: very small) , accounting for 40.76% of all bad credits.

FIFTH QUERY:



From this query, we can conclude that the average loan amount for **bad credits** is €486.81 higher than that for **good credits**.

STATISTICAL TESTS

T-TEST: The first test conducted was a t-test (t-test sheet) in order to assess whether the mean loan amount differs significantly across default status groups. To perform this test, **the null hypothesis (H_0) was defined as there being no statistically significant difference in the average loan amount**

(amount_euro) between applicants classified as having good credit and those classified as having bad credit. The alternative hypothesis (H_1) was defined as there being a statistically significant difference in the average loan amount (amount_euro) between these two groups.

The next step was to set the **Significance level at 0.05** (confidence level 95%), which is commonly used as a threshold for hypothesis testing, and to calculate the p-value. **The p-value indicates the probability that the observed result could occur by chance.** In this case, the **p-value was equal to 0.00002477691431.**

According to hypothesis testing theory, if the p-value is less than the significance level, we reject the null hypothesis (H_0) and accept the alternative hypothesis (H_1). ***Therefore, there is a statistically significant difference in the average loan amount (amount_euro) between applicants classified as having good credit and those classified as having bad credit.***

In order to investigate whether the difference between the average loan amounts of bad and good credits is practically significant, an additional calculation was required. The **Welch-type standardized difference (d)**, an alternative to **Cohen's d formula**, indicates the size of the practical effect. In our case, **the d value was equal to 0.32.** According to Cohen's d theory, this value indicates a small practical difference between the two averages. ***This indicates a small association between loan amount and default status.***

CHI-SQUARE TEST: The second test conducted was a chi-square test (chi-square test sheet) in order to investigate whether there is a statistically significant difference between the installment rate categories and the default status. To perform this test, **the null hypothesis (H_0) was defined as there being no statistically significant difference between the installment rate categories and the default status. The alternative hypothesis (H_1) was defined as there being a statistically significant difference between the installment rate categories and the default status.**

The next steps were the same as those mentioned in the t-test. First, **the significance level was set at 0.05 (confidence level 95%)**, and then the p-value was calculated. In this case, the **p-value was equal to 0.06989890753**, which means that ***there is no statistically significant difference between the installment rate categories and the default status.***

EXECUTIVE SUMMARY

- *The risk is considered higher for applicants who are young and also unemployed or have a very short employment history.*
- *Banks should be more careful before approving loans for applicants who belong to these categories. It may be better to ask for loan guarantors, real property, mortgages, etc.*

- *Banks could approve smaller loan amounts or impose higher interest rates for these categories of applicants.*